Question	Answer	Marks
1(a)(i)	direction of the force acting on a (test) mass placed at the point	**B1**
1(a)(ii)	change in height negligible compared with radius (of Earth)	**B1**
	(so) field lines are (effectively) parallel	**B1**
1(b)(i)	$Y = GM / R^2$	**M1**
	G is the gravitational constant	**A1**
1(b)(ii)	gravitational force is (always) attractive **or** gravitational force (always) acts towards the centre of the sphere	**B1**
	force is in opposite direction to displacement **or** at a point to the right of the centre, force acts to the left **or** at a point to the left of the centre, force acts to the right	**B1**
1(b)(iii)	sketch: smooth curve with decreasing positive gradient, starting at (R, $-Y$) and reaching $3R$ with g still negative **or** smooth curve with increasing positive gradient, ending at ($-R$, Y) and reaching $-3R$ with g still positive	**B1**
	both of the above curves, in correct quadrants	**B1**
	curve passing through ($\pm 2R$, $\pm 0.25Y$) and ($\pm 3R$, $\pm 0.11Y$)	**B1**

Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to change temperature)	B1